Paleomagnetism of the Little Dal lavas, Mackenzie Mountains, Northwest Territories, Canada

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The Proterozoic stratigraphic column of the Mackenzie Mountains is dominated by two main successions, the platformal "Mackenzie Mountains Supergroup" beneath, and the Windermere-equivalent Rapitan Group and younger strata, of rift-depression and slope origin, above. The former succession is at least partly older than 770 Ma, the latter younger than 770 Ma. These two main successions are locally separated by an unconformity-bounded succession, the "copper cycle." An important question is whether the copper cycle is more closely related in time and in origin to the older or the younger main succession. Determination of the paleomagnetism of the basaltic lavas locally preserved at the top of the Little Dal Group (top of the Mackenzie Mountains Supergroup) and comparison of their remanence directions with those published for other rocks bearing on the question were thought to be one way of shedding light on this question. Accordingly, paleomagnetic investigation of 10 sites in the Keele River area and six sites in the Thundercloud Range area was undertaken to obtain the remanence direction related to the initial extrusion of the lavas. Coherent directional groupings were only obtained from Little Dal lavas in the Keele River area. Of the three magnetizations found, LD-L ($D = 304°$, $I = 20°$, $\alpha_{95} = 7°$) is assumed to represent the magnetization acquired on crystallization of the lavas. If this assumption is correct, the significant difference from the direction LD-A ($D = 265°$, $I = 26°$, $\alpha_{95} = 4°$) reported elsewhere for strata low in the Little Dal Group suggests either that the lavas significantly postdate the group or that significant movement of the North American plate occurred during accumulation of the 2 km or so of platformal strata between the lower Little Dal beds and the lavas. The new results presented here also admit the conclusion that the Little Dal lavas do not represent the same igneous events as diabase intrusions dated at about 770 Ma that cut the lower Little Dal.

La colonne stratigraphique protérozoïque des monts Mackenzie est composée en prédominance de deux successions principales, en dessous se trouve le "supergroupe des monts Mackenzie" originant sur la plate-forme, et au-dessus on observe le groupe Rapitan, équivalent au Windermere, et des strates plus jeunes, originaires d'un fossé d'effondrement et de la pente continentale. La première succession est au moins en partie plus âgée que 770 Ma, la dernière est plus récente que 770 Ma. Ces deux successions principales sont séparées localement par une discordance liée à une succession, "le cycle de cuivre." Une question importante est à savoir si le cycle de cuivre est plus étroitement relié, dans le temps et par l'origine, avec la succession principale la plus ancienne ou avec la plus récente. La détermination du paléomagnétisme des laves basaltiques préservées localement au sommet du groupe Little Dal (sommet du supergroupe des monts Mackenzie), et une comparaison de leur direction de rémanence avec celles publiées antérieurement pour d'autres roches apparentées à ce problème, ont été considérées comme un moyen d'apporter des éléments de réponse à la question. En conséquence, une étude paléomagnétique sur 10 sites dans la région de la rivière Keele et sur six sites dans la région de Thundercloud Range a été entreprise pour déterminer la direction de rémanence liée à l'extrusion de la lave. Une direction cohérente d'un groupe d'échantillons a été obtenue pour les laves de Little Dal dans la région de la rivière Keele. Parmi les trois directions d'aimantation mesurées, LD-L ($D = 304°$, $I = 20°$, $\alpha_{95} = 7°$) est considérée comme étant l'aimantation acquise lors de la cristallisation des laves. Si ce résultat est supposément correct, la différence significative d'avec la direction LD-A ($D = 265°$, $I = -26°$, $\alpha_{95} = 4°$) rapportée ailleurs pour les couches en position inférieure dans le groupe Little Dal indique soit que les laves sont définitivement postérieures au groupe, ou qu'un déplacement significatif de la plaque de l'Amérique du Nord se produisit durant la période d'accumulation d'environ 2 km de couches sur la plate-forme entre les lits de la zone inférieure du groupe Little Dal et les laves. Les nouveaux résultats sont compatibles avec la conclusion que les laves de Little Dal ne correspondent pas au même phénomène d'intrusion des diabases datées à environ 770 Ma, lesquelles recoupent la partie inférieure du groupe Little Dal.

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Introduction

The Proterozoic column of the Mackenzie Mountains is dominated by two major sequences (Fig. 1). The older "Mackenzie Mountains Supergroup" is characterized by chemical and mature clastic strata; it comprises all formations from "Map-unit H1" (the oldest exposed) to

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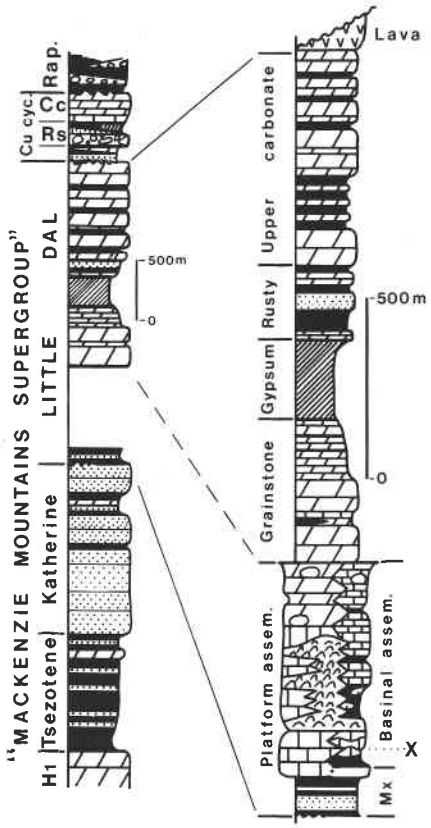


FIG. 1. Stratigraphy of the "Mackenzie Mountains Supergroup" and the Little Dal Group (diagrammatic). Cu cyc. = "copper cycle"; RS = Redstone River Formation; Cc = Coppercap Formation; Rap. = Rapitan Group; X = beds sampled by Park (1981*a*).

the top of the Little Dal Group (Aitken 1981; Aitken *et al.* 1982). The younger major sequence is dominated by clastic strata that grade from immature below to mature above; it extends from the base of the Rapitan Group to the sub-Cambrian unconformity, and has been correlated with the Windermere Supergroup of the southern Canadian Cordillera (Gabrielse 1972; Gabrielse *et al.* 1973; Eisbacher 1978, 1981; Young *et al.* 1979).

The two major sequences are separated in some areas by an unconformity-bounded sequence, the "copper cycle" (Aitken 1981), comprising the Redstone River and Coppercap Formations and some underlying strata formerly assigned to the upper Little Dal (Fig. 1).

Present knowledge of the Proterozoic of the Mackenzie Mountains leaves several important questions unanswered:

(1) Is the copper cycle more closely related to the overlying Windermere-equivalent sequence, or to the underlying Mackenzie Mountains Supergroup, within which it has been included by most writers? In other words, which unconformity represents the greater hiatus, the locally spectacular sub-Rapitan, or the more subtle sub-copper cycle?

(2) The base of the lavas at the top of the Little Dal Group appears conformable, whereas their top is an erosional surface. On the other hand, copper cycle deposition has been interpreted as a response to faulting (Eisbacher 1977; Jefferson 1978). Are the "Little Dal lavas" so close in age to the copper cycle that they, too, could be an early response to the same faulting event?

(3) Do the diabases (dated at about 770 Ma by Armstrong *et al.* 1982) that intrude the lower Little Dal and contribute clasts to the Shezal Formation (middle Rapitan) belong to the same igneous event as the Little Dal lavas?

Lithostratigraphic analysis has been carried almost to its limits in attempts to answer these questions, and rocks suitable for radiometric dating have not been found. Studies of the paleomagnetism of the strata concerned provide one possible means of approaching these problems; this report offers new paleomagnetic data on the Little Dal lavas and examines the significance of these data in relation to paleomagnetic data published or available for other rocks involved in the questions, namely, near-basal Little Dal sedimentary rocks (Park 1981*a*) and the diabases intrusive into the lower Little Dal (Park 1981*b*; Morris and Park 1981). The assumption made throughout is that rock units with similar magnetic signatures may be of similar age, while divergent magnetic signatures suggest a significant difference in age.

Methods

The Little Dal lavas were sampled at two localities separated by approximately 150 km. From two sections through the lava sequence near the Keele River area (63°48′N, 231°30′E) 10 sites were obtained for paleomagnetic analysis. Six sites were sampled at a second locality near Coates Lake in the Thundercloud Range (62°24′N, 233°40′E). At each site at least six 2 in. (5 cm) cores were collected by field drilling. Two specimens were sliced from each core. The bedding orientation at each sampling locality was measured on the nearest sedimentary unit. The difference in structural orientation between the two areas provided a possible fold test of remanence age. At the Keele River sites the beds dip moderately to the west, and at the Coates Lake sites the beds dip steeply to the east. As this folding was probably developed during the Mesozoic orogeny, the fold test does not provide a precise age control of the magnetization; magnetizations passing a positive fold test are merely constrained to the interval Late Precambrian through Cretaceous.

TABLE 1. Summary of determinations of stable magnetizations LD-N, LD-M, and LD-L, for sites in the Keele River area

	LD-N[a]							LD-M[b]								LD-L[c]					
					In situ		Tilt-corrected				*In situ*		Tilt-corrected					*In situ*		Tilt-corrected	
Site	*N*	*R*	*k*	α_{95}°	*D*°	*I*°	*D*°	*I*°	*N*	*R*	*k*	α_{95}°	*D*°	*I*°	*N*	*R*	*k*	α_{95}	*D*°	*I*°	*D*°	*I*°
12	5	4.897	39	12	003	70	289	73	6	5.964	137	6	318	53	290	46	6	5.951	103	7	318	24	307	22
13	11	10.954	216	3	014	74	273	77	3	2.989	241	8	272	75	246	54								
14	21	20.848	132	3	352	78	262	72	15	14.825	80	4	297	58	272	45								
15	6	5.971	174	5	008	83	247	71	5	4.914	47	11	285	60	264	44								
16	4	3.973	113	9	324	79	256	66	5	4.974	151	6	281	66	257	47								
38													(318 | 56 | 287 | 50)	9	8.912	92	5	315	31	300	26
37	3	2.964	55	18	347	66	292	65								11		10.861	72	5	311	20	304	15
36					(339 | 64 | 293 | 64)								10		9.903	93	5	312	20	305	15
35	6	5.938	81	8	356	75	67	70																
34	6	5.959	122	6	354	73	271	66																

[a]Mean direction: *in situ*: $D = 355°$, $I = 75°$, pole lat. = 87°N, long. = 104°E; tilt-corrected: $D = 270°$, $I = 71°$, pole lat. = 47°N, long. = 175°E. $N = 8$ sites; $R = 7.955$; $k = 159$; $\alpha_{95} = 4°$.
[b]Mean direction: *in situ*: $D = 294°$, $I = 63°$, pole lat. = 50°N, long. = 147°E; tilt-corrected: $D = 266°$, $I = 48°$, pole lat. = 24°N, long. = 159°E. $N = 5$ sites; $R = 4.918$; $k = 49$; $\alpha_{95} = 11°$.
[c]Mean direction: *in situ*: $D = 314°$, $I = 24°$, pole lat. = 29°N, long. = 106°E; tilt-corrected: $D = 304°$, $I = 20°$, pole lat. = 24°N, long. = 115°E. $N = 4$ sites; $R = 3.984$; $k = 186$; $\alpha_{95} = 7°$.

Each specimen was individually demagnetized by either thermal or alternating field (AF) treatment. All measurements and remanence treatments were performed on Schonstedt equipment (DSM, TSD, GSD). For each specimen, stereonets, intensity decay plots, and orthogonal diagrams were plotted to aid recognition of the individual remanence directions and their magnetic characteristics.

Individual remanence components were initially identified by a visual inspection of the orthogonal diagrams. Numerical estimates of the direction of each component were obtained by one of three methods. The reference technique was the least-squares fitting of lines to segments of demagnetization trajectory (Kirschvink 1980). Minimum acceptance criteria adopted for this technique were: a minimum of four points on a line, with a maximum angular deviation of 7.5°. Other techniques included least-squares fits to great-circle distributions (Halls 1976), and sequential vector subtractions. For these techniques, which only provided supporting evidence, the minimum acceptance criteria were, respectively: a minimum of four great-circle segments with arc lengths greater than 25°, required to define an overprint component; and a maximum angular difference of 10° (determined by vector subtraction) between three sequentially subtracted vectors from the same specimen, the accepted direction being a vector mean of the three components. Site mean directions representative of the individual remanence components present at each site were calculated using the standard Fisher (1953) statistical techniques or the Kirschvink-derived (1980) specimen data directions.

Results

Keele River

At each of the two sections examined, the lava pile could be subdivided into two distinct portions: a lower portion of relatively unaltered lava, and an upper portion that exhibits the effects of extensive subaerial weathering. In this upper portion, red blebs and veins of jasper are common. This feature is reflected to some degree by the variations in natural remanent (NRM) intensity. In the least altered, lower portion of the lava pile intensities up to 1 A m^{-1} are observed, whereas in the upper, most altered portions the intensity is reduced to levels of around 9–10 mA m^{-1}. All specimens have a significant portion of their remanence residing in hematite. The average acquisition coercive force for these specimens is around 70–80 mT but in some specimens it is found to range up to 150 mT.

From these specimens it was possible to obtain mean directional estimates for three distinct remanence vectors (Table 1). Two other vectors found in a few specimens could not be satisfactorily resolved by any of the analyical techniques, partly because determinations

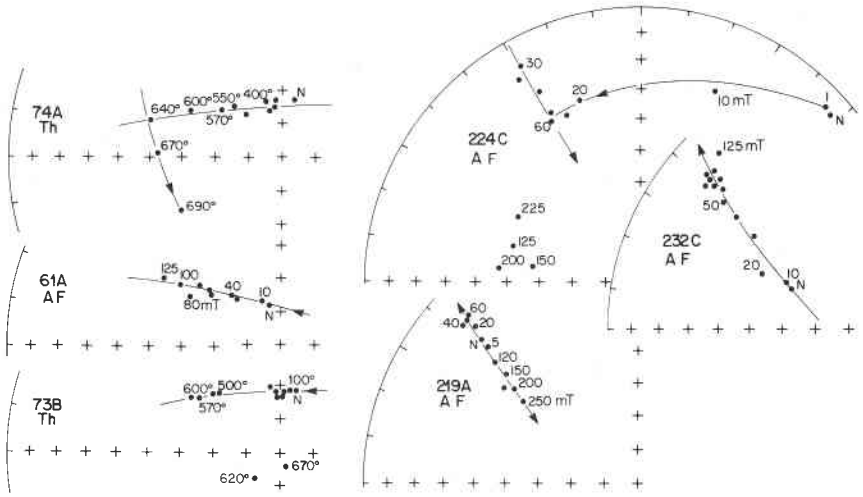


FIG. 2. Equal-area stereographic projections showing the change of *in situ* remanence direction upon alternating field and thermal demagnetizations of selected specimens from the Keele River area.

were too few to provide any meaningful statistical estimates among sites.

Direction LD-L (four sites; $D = 304°$, $I = 20°$, $\alpha_{95} = 7°$) possibly represents the earliest phase of magnetization found at this locality. LD-L is predominantly found in the lowermost, least altered parts of the lava pile. As shown by Fig. 2 it commonly has intermediate resistive coercive force (RCF), which is isolated most often in the range 30–60 mT. Occasionally this direction is retained up to alternating fields as high as 125 mT (232C, Fig. 2). Further treatment beyond this point either produces a sudden shift to a different remanence vector (224C, Fig. 2) or a complete loss of directional coherence. Upon thermal demagnetization, remanence unblocking is found over a limited temperature range; for hematite this is around 660°C, and for magnetite around 570°C. The relative proportions of hematite to magnetite varies from specimen to specimen even within the area of a single sampling site. LD-L is the only direction observed in univectorial samples. All other remanence directions are found as a result of the resolution of multicomponent demagnetization trajectories.

Directions LD-M ($N = 5$ sites, $D = 266°$, $I = +48°$, $\alpha_{95} = 11°$) and LD-N ($N = 8$ sites, $D = 270°$, $I = +71°$, $\alpha_{95} = 4°$) are found predominantly in the upper, more hematitic portions of the lava pile. LD-M most definitely resides in hematite (Table 1). Upon AF demagnetization, specimens that initially show the LD-L direction often exhibit the LD-M direction after demagnetization beyond 125 mT. Figure 2 shows two examples (224C, 219A) where the demagnetization sequence is LD-L→ LD-M. Similarly, AF demagnetization of LD-M + LD-N specimens always shows a progression to the LD-M vector in higher alternating fields. Thermal demagnetization experiments further confirm this observation. Examples 74A and 73B (Fig. 2) both show a shift towards the LD-M direction from LD-N, especially after thermal demagnetization up to temperatures where only magnetizations residing in hematite are detected.

The magnetic mineralogy corresponding to the LD-N direction is less certain. From the observations described above, it clearly has generally lower coercivity and unblocking temperatures than the LD-M direction residing in hematite. It is unlikely that the remanence carrier is magnetite because, as mentioned above, the hematite-rich specimens carrying the LD-M and LD-N vectors all have substantially reduced NRM intensities and upon thermal demagnetization there is no pronounced remanence unblocking at 580°C. Typically, LD-N is characterized by unblocking temperatures at <400°C, and resistive coercive forces of less than 20 mT. A possible carrier of the LD-N remanence is goethite; this may have been produced by recent hydration of the hematite-rich lavas.

Of the two unresolved vectors, found in a few specimens, the most common one has north-northeast declination and a shallow positive inclination. Figure 3 shows a plot of the great-circle intersections that provide a directional estimate of this vector. All examples of this component show low remanence stability. Upon AF treatment the vector is always removed by 20 mT (224C, Fig. 2), whereas upon thermal treatment the vector is never found to survive temperatures in excess of 300°C. The least common vector in this data set has an easterly declination and shallow negative inclination. This vector is observed in only eight specimens, all of which are from the more hematitic parts of the lava pile.

There are no conclusive tests to establish the relative ages of any of these magnetizations. In the absence of

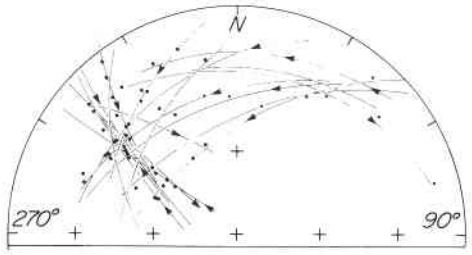


FIG. 3. Great-circle demagnetization trajectories of tilt-corrected remanence directions from specimens of the Keele River area. Two regions of intersection between the great circles can be recognized: shallow positive inclination to the northeast, and shallow intermediate inclination to the northwest. Equal-area stereographic projection.

any evidence to the contrary, we assume that the association of LD-L with the unaltered lava may be taken as indicating that LD-L may be an initial magnetization. Certainly the fact that the hematite and magnetite estimates of LD-L vector are in close agreement supports this suggestion. LD-M appears to be intimately associated with the hematization of the upper part of the lava pile. The presence of pebbles of jasper in a conglomerate below the immediately overlying Redstone River Formation suggests the hematization was pre-Redstone. The age of the LD-N direction is the least well constrained. Comparing a similar pole from low in the Little Dal Group to the Phanerozoic pole path for North America, Park (1981*a*) preferred a Cretaceous age for this remanence, associating its acquisition with Cretaceous orogeny in the Mackenzie Mountains. Its possible age range is Late Precambrian to Cretaceous.

Thundercloud Range

Of the six sites sampled in the Thundercloud Range, four sites (1–4) are at progressively lower stratigraphic levels in the lava pile. Average spacing between the sites is approximately 5 m. All these sites are in massive amygdaloidal lava units. Sites 5 and 6 are in a plug that outcrops immediately below, and is a possible feeder of the lavas. Specular hematite occurs along fractures within the lava, and defines the weathering surfaces of individual flow tops. Natural remanent intensities at these sites are significantly higher than at the Keele River sites, ranging from 6 to 100 A m^{-1}. Median destructive fields for these specimens fall into two groups: 5 mT, where magnetite remanences predominate, and 50 mT, where hematite remanences are more important. The individual unblocking temperatures (T_{ub}) are somewhat variable. Most specimens exhibit a T_{ub} point in the range 450–500°C. The temperature range over which this unblocking occurs is 50–80°C. Other important unblocking temperature points are found around 580 and 630°C; these have much narrower thresholds, usually 5–10°C. Exceptions to these

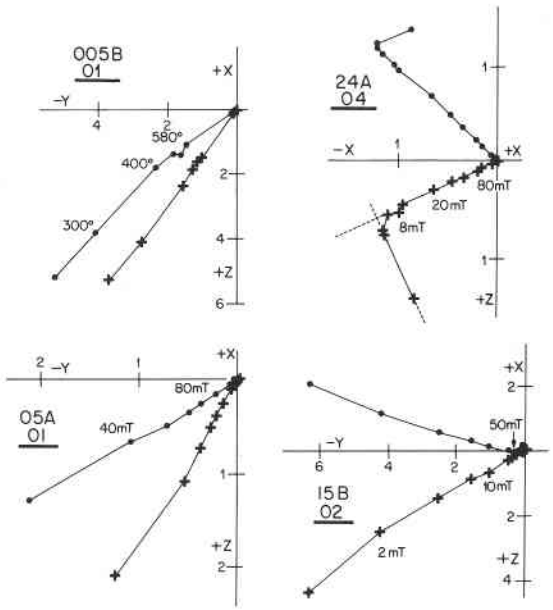


FIG. 4. Examples of the demagnetization characteristics of samples from the Thundercloud Range. All samples have tilt-corrected southwesterly oriented declinations, but these are found with both positive and negative inclinations. Orthogonal vector projections: crosses represent projections onto the horizontal (*XY*) plane, and dots represent projections onto either of the two vertical planes (*ZX*, *ZY*). The adopted convention is: $+X$ is north, $+Y$ is east, and $+Z$ is down.

generalizations are a number of specimens from sites 5 and 6. In all cases, these are characterized by low unblocking temperatures (<350°C) and low resistive coercive forces (directions are retained only to 15–20 mT).

Only site 4 shows a well defined mean grouping of remanence directions ($D = 198°$, $I = -29°$, $\alpha_{95} = 12°$). All other sites show a broad smearing of remanence directions. The greatest concentration has very shallow inclination and is smeared across the southwest quadrant (Fig. 5). Although the data for each site are characterized by smearing, individual specimens give very well defined straight-line trajectories that decay to the origin (Fig. 4). Adjacent specimens from the same core most often have similar but not necessarily identical remanence directions (compare 5B 01 and 5A 01, Fig. 4). Two possible mechanisms may have caused the smearing. The individual observed vector may be the resultant of two (or more) actual vectors that have similar remanence characteristics such that any treatment produces straight-line vector trajectories from which the individual components cannot be separated. On the oher hand, each direction may be a real record of the Earth's field direction, in which case the magnetic component is secondary and was acquired over a

geologically significant period of time. Because the lavas and plug at this locality are extensively altered, without conclusive evidence to the contrary the latter alternative is preferred.

Discussion

A discussion of the significance of the new data begins with a comparison between the initial remanence directions for the Little Dal lavas and those established for the lower Little Dal Group by Park (1981*a*). Park reported three directions from sedimentary rocks of the "basinal sequence" of the Little Dal. The magnetite- and hematite-residing LD-A direction is a pre-folding magnetization (pre-Late Cretaceous) which has a mean direction of $D = 265°$, $I = -26°$ ($\alpha_{95} = 4°$). A second magnetization, LD-B, thought to be residing in the red hematite pigment, has a similar mean direction at $D = 273°$, $I = -09°$ ($\alpha_{95} = 10°$). Park (1981*a*) considered that both these magnetizations were acquired "soon after rock genesis." A third direction, LD-C, ($D = 289°$, $I = +79°$, $\alpha_{95} = 7°$), which characteristically has low blocking temperatures, was suggested to be a "goethite" weathering component of Cretaceous age.

The lavas have a much more complicated remanence history. There seems to be little obvious correlation between the remanence directions from the two sections at which the lavas were sampled. At the Keele River area, three well defined groups of remanence directions could be separated. Because of its occurrence in the unaltered lava, and as an univectorial magnetization that resides in both magnetite and hematite, the direction LD-L is considered to provide the best estimate of the magnetic field direction during lava extrusion for the Keele River area. This direction is strongly divergent from the LD-A direction established by Park (1981*a*) for the lower Little Dal Group. Available evidence does not permit a choice between the two obvious hypotheses for explaining this divergence: (a) LD-A is a direction representative of the Little Dal Group as a whole; the divergence of the LD-L direction is evidence for a significant hiatus between the dates of deposition of the upper carbonate formation of the Little Dal and the Little Dal lavas. (b) The 2 km, more or less, of platformal sedimentary rock between the strata sampled by Park (1981*a*) and the top of the Little Dal upper carbonate accumulated over a long period of time, during which the North American plate moved 27° latitudinally (the calculated angular difference between LD-A and LD-L). Thorough paleomagnetic sampling of the Little Dal Group would probably yield data upon which a choice between these two hypotheses could be made.

The LD-M direction resides mainly in hematite. The hematitic alteration and jasper veining of the lavas are spatially related to the erosional contact at the top of the lavas; furthermore, clasts of jasper and pebbles with jasperized rims are present in overlying conglomerates. Clearly, the remanence direction residing in hematite, namely, LD-M, was acquired during the post-lava, pre-copper cycle interval of erosion and weathering. The angular difference of 15° of latititude between LD-L and LD-M implies that a significant period of time intervened between eruption of the lavas and the hematite–jasper weathering event; accordingly, the tentative decision to include the lavas within the copper cycle (Aitken 1981) should be reversed.

No overall directional groups were resolved for the sites in the Thundercloud Range area. A crude mean of the broad smear of results appears to be somewhat similar (but antipolar) to the shallow northwesterly directions that are found in some samples from the Keele River section. Some of the directions are also broadly similar to the LD-B direction that Park (1981*a*) relates to hematite staining in the Little Dal Group sediments (Fig. 5*a*). As suggested above, it is possible that many of the directions from the Little Dal lavas in the Thundercloud Range are also secondary magnetizations related to hematite staining. While LD-B records only one phase of remanence blocking, the Thundercloud Range samples may be showing the effects of blocking over a much longer period of time.

Lack of agreement between the remanence directions residing in hematite in the lavas sampled at Keele River and those sampled in the Thundercloud Range might be taken as evidence that the lavas at the two localities were erupted at different times. This appears implausible, however, because the position of the lavas, not only with respect to underlying units of the upper Little Dal Group, but also with respect to copper cycle beds overlying the unconformity, is identical at the two localities (Aitken 1981).

At present, the best estimates of the original magnetizations of the lower Little Dal Group sediments and the Little Dal lavas appear to be, respectively, LD-A ($D = 265°$, $I = -26°$) and LD-L ($D = 304°$, $I = +20°$). A significant portion of the difference between these two directions is in magnetic inclination (Fig. 5). Hence, unless one argues for geologically unreasonable rotations it is not possible to explain this directional discrepancy by differential rotation on thrust sheets.

A further conclusion can be drawn by comparing the remanence directions of the lavas with those determined by Park (1981*b*) for the diabase intrusions that cut the Mackenzie Mountains Supergroup as high as the lower Little Dal (see Morris and Park 1981). Armstrong *et al.* (1982) report Rb–Sr isochron dates of 766 ± 24 and 769 ± 27 Ma for the intrusions sampled. The significant difference between the remanence directions of the Little Dal lavas and the diabase intrusions ($D = 272°$, $I = 00°$, $\alpha_{95} = 6°$) implies that the lavas and the intrusions represent two separate igneous events, rather than

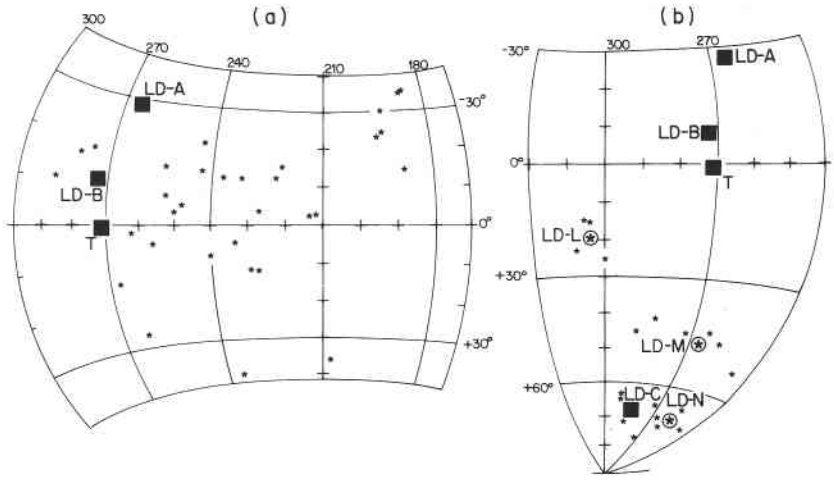


FIG. 5. Comparisons of individual specimen directions (asterisks) from the the Thundercloud Range (*a*) and site mean directions (LD-L, LD-M, LD-N) from the Keele River area (*b*), with the directions LD-A, LD-B, LD-C reported by Park (1981*a*) from the Little Dal Group sediments, and the direction *T* from sills intrusive into the Tsezotene Formation. Equatorial equal-area stereographic projection; all directions tilt-corrected.

a single event, as some have considered possible (e.g., Armstrong *et al.* 1982). Unfortunately, it is not presently possible to state on the basis of paleomagnetism whether the lavas are older or younger than the dated intrusions.

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AITKEN, J. D. 1981. Stratigraphy and sedimentology of the Upper Proterozoic Little Dal Group, Mackenzie Mountains, N.W.T. *In* Proterozoic basins in Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 47–71.

AITKEN, J. D., COOK, D. G., and YORATH, C. J. 1982. Upper Ramparts River (106 G) and Sans Sault Rapids (106 H) map-areas, District of Mackenzie. Geological Survey of Canada, Memoir 388.

ARMSTRONG, R. L., EISBACHER, G. H., and EVANS, P. D. 1982. Age and stratigraphic–tectonic significance of Proterozoic diabase sheets, Mackenzie Mountains, northwestern Canada. Canadian Journal of Earth Sciences, **19**, pp. 316–323.

EISBACHER, G. H. 1977. Tectono-stratigraphic framework of the Redstone Copper Belt, District of Mackenzie. *In* Report of activities, part A. Geological Survey of Canada, Paper 77-1A, pp. 229–234.

——— 1978. Re-definition and subdivision of the Rapitan Group, Mackenzie Mountains. Geological Survey of Canada, Paper 77-35.

——— 1981. Sedimentary tectonics and glacial record in the Windermere Supergroup, Mackenzie Mountains, northwestern Canada. Geological Survey of Canada, Paper 80-27.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**, pp. 295–305.

GABRIELSE, H. 1972. Younger Precambrian of the Canadian Cordillera. American Journal of Science, **272**, pp. 521–536.

GABRIELSE, H., BLUSSON, S. L., and RODDICK, J. A. 1973. Geology of Flat River, Glacier Lake, and Wrigley Lake map-areas, District of Mackenzie and Yukon Territory. Geological Survey of Canada, Memoir 366.

HALLS, H. C. 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**, pp. 297–304.

JEFFERSON, C. W. 1978. Stratigraphy and sedimentology, Upper Proterozoic Redstone Copper Belt, Mackenzie Mountains, N.W.T.—a preliminary report. Department of Indian and Northern Affairs, Canada, Mineral Industry Report, 1975, Northwest Territories EGS 1978–5.

KIRSCHVINK, J. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**, pp. 699–718.

MORRIS, W. A., and PARK, J. K. 1981. Correlation of Upper

Proterozoic strata in the Cordillera: paleomagnetism of the Tsezotene sills and the Little Dal lavas. *In* Proterozoic basins in Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 73–78.

PARK, J. K. 1981*a*. Analysis of the multicomponent magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada. Journal of Geophysical Research, **86**, pp. 5134–5146.

——— 1981*b*. Paleomagnetism of the sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **18**, pp. 1572–1580.

YOUNG, G. M., JEFFERSON, C. W., DELANEY, G. D., and YEO, G. M. 1979. Middle and Late Proterozoic evolution of the north Canadian Cordillera and Shield. Geology, **7**, pp. 125–128.